Use LLMs to Build Repair Tools, Not Patch Generation: Cost-Effective Automated Program Repair

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Abstract

Automated program repair can give every student in a programming course immediate feedback. Reference-guided repair transplants from the correct programs a class has already submitted and costs nothing per submission, but fails where the pool holds no structurally matching program. LLM-based repair has no such limit, yet it issues one query per submission, so spending grows with enrollment and recurs every term while the queries to one assignment re-read the same specification and re-derive the same few misconceptions hundreds of times. We present CoEP, in which the model writes the repair *tool* rather than the patch. Once per assignment, a builder LLM turns worked pairs drawn from the class's own submissions into a module of misconception-named rewrite rules over a reference-guided transplant, validated against the instructor's test cases and then frozen. Submissions are repaired by that module with no model at all, and only what it abstains on, or what it patches more heavily than the program is long, takes a conventional call; the repairs it cannot make are distilled back into rules it keeps, so the artifact develops with the assignment it serves. What is deployed is not a model but code that can be read, audited, and that names the misconception behind every repair.

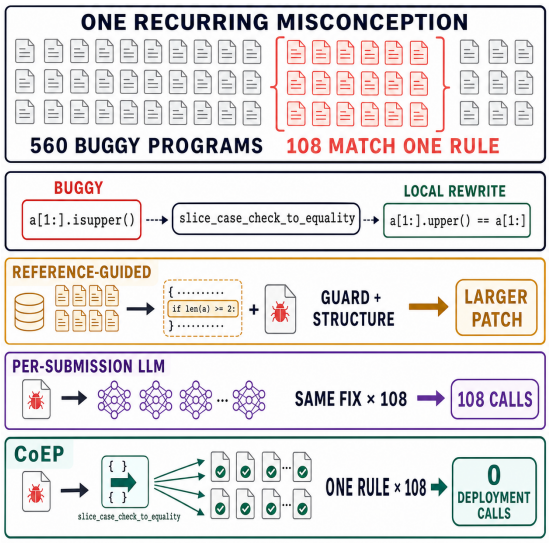


Figure 1: Example of CoEP: A recurring misconception on cAPS lOCK and the rule that repairs it.

Introduction

Every year, thousands of students enroll in programming courses, and online judges accumulate millions of submissions to the same set of problems. With the rapid growth of Computer Science enrollment, providing personalized and timely feedback on programming assignments has become a significant challenge, requiring substantial time and effort from faculty (Yi et al. 2017; Hu et al. 2020).

Automated Program Repair (APR) has emerged as a promising solution to this challenge, aiming to deliver automated and personalized feedback to students about their programming errors (Gulwani, Radiček, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2020; Ahmed et al. 2022). Reference-guided APR aligns a buggy program with a correct program for the same problem and transplants the differing region. It uses no model at deployment, so it runs offline and deterministically at no marginal cost per submission. Its applicability depends on the pool of correct programs containing a structurally matching correct program: when the alignment is not possible, the tool reports a structural mismatch and the program is returned unrepaired (Orvalho, Janota, and Manquinho 2025).

More recently, Large Language Models (LLMs) trained on code have shown great potential in repairing programs (Xia and Zhang 2022, 2024; Zhang et al. 2024; Liu et al. 2024; Yang et al. 2024). LLM-based APR does not require a matching reference, so it applies where a structural mismatch would otherwise arise, and it extends beyond introductory exercises to harder problems. What concerns us is how it is deployed. The default is one model query per buggy program, so spending, latency, and dependence on an external service all scale with the number of submissions and recur every term. Queries to the same assignment also share most of their content: hundreds of students attempt one problem and their failures cluster into a few recurring misconceptions (Gulwani, Radiček, and Zuleger 2018; Hu et al. 2020), so the model re-reads one specification and re-derives one mistake many times over. Cheaper tiers and prompt caching lower the price of a call (Anthropic 2025; OpenAI 2024) but leave the number of calls unchanged.

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In this paper, we propose CoEP (Cost-Effective Program repair), in which the model writes the repair *tool* rather than the patch generation. The model is invoked once per problem instead of once per submission, and the product of that invocation is not a single repair but an executable program that handles every submission to the problem. An LLM agent receives the problem description, its test cases, and a clustered sample of (buggy, fixed) program pairs drawn from the class, and authors AST rewrite rules named after the misconceptions they repair. These rules run atop a shared symbolic substrate of a generic edit library and reference-guided minimal-hunk transplant from the pool of correct programs. Figure 1 shows one such rule. On cAPS lOCK, 108 of 560 buggy programs test a word's tail with `a[1:].isupper()`, which is `False` on the empty tail of an input like `"Z"`. Repairing them one at a time re-derives the same fix 108 times, and a transplant must bring in a length guard with the structure around it. One rule instead rewrites that test in place, repairs all 108, of which the builder was shown 64 and never saw the other 44, at no deployment cost. Every candidate is validated against the instructor's test cases before it is kept, so a defective tool abstains rather than answers wrongly, and what it abstains on escalates to a conventional call, so coverage is never sacrificed. Model spend moves from a marginal cost that scales with submissions to a fixed one that scales with problems, and what a student meets is code that can be read and audited, running inside the institution.

Our experiments with 2,970 buggy programs across 11 problems from a CS1 corpus and an online judge reveal that the synthesized tool outperforms Refactory, the strongest reference-guided repairer that runs on our data, on its own corpus and metric, repairing more programs with smaller patches. Against per-submission LLM repair it matches coverage while modifying less of the student's program, for a fraction of the model spend, and it recovers its one-off build cost within tens of submissions per problem. An ablation separates the layers. Where assignments already carry hundreds of correct submissions the reach is the substrate's and the rules add smaller patches and a misconception label; restrict the pool and the balance inverts, the substrate repairing 11.6% of held-out submissions where the same tool with its rules repairs 32.2%. Those two are what a per-submission call cannot amortize, since a call yields a patch but leaves no reusable artifact behind.

In summary, this paper makes the following contributions:

- We reformulate APR for programming assignments so that the model authors a repair tool once per assignment rather than a patch per submission, leaving no model in the deployed system;
- We present CoEP, which synthesizes misconception-named rewrite rules atop a reference-guided transplant, and keeps developing the tool by distilling the repairs it could not make into new rules;
- Experiments on real student submissions show that CoEP repairs more programs with smaller patches than a reference-guided tool, and matches per-submission LLM repair at an order of magnitude lower cost.

Related Work

APR for programming assignments. Early work reuses the correct programs a class has already submitted. CLARA and Sarfgen align a buggy program with its structurally nearest correct program and transplant the differing blocks (Gulwani, Radiček, and Zuleger 2018; Wang, Singh, and Su 2018), Verifix backs that alignment with formal verification (Ahmed et al. 2022), and Refactory widens the range of alignable programs by refactoring the pool to match the buggy program's structure (Hu et al. 2020). A second line learns the repair function from data: sequence models (Gupta et al. 2017), reinforcement learning over edits (Gupta, Kanade, and Shevade 2019), and PROSE-based induction of per-assignment transformations (Rolim et al. 2017). Most recently, LLMs trained on code address the problem directly: PyDex targets student programs and prepends similar correct programs retrieved by test-case behaviour as few-shot examples (Zhang et al. 2024), ChatRepair adds multi-turn conversation (Xia and Zhang 2024), CREF places a tutor in the loop (Yang et al. 2024), FastFixer fine-tunes for faster per-query inference (Liu et al. 2024), and recent work extends to counterexample-guided minimal repair (Orvalho, Janota, and Manquinho 2025) and learner-tailored retrieval (Dai et al. 2026). Alongside repair itself, OverCode clusters a class's programs to show an instructor which mistakes recurred (Glassman et al. 2015), the same instructor-facing view our rule provenance yields as a by-product of repairing. **Letting a model build the tool.** The reformulation this paper applies to repair has been made elsewhere: a capable model writes a reusable utility once and cheaper calls reuse it (Cai et al. 2023), skills accumulate from experience (Wang et al. 2023), and functions are induced from programmatic tasks or evolved against a score (Wang, Fried, and Neubig 2024; Stengel-Eskin, Prasad, and Bansal 2024; Romera-Paredes et al. 2024). What repair contributes is an oracle. A tool induced for reasoning is trusted because it looks right, whereas every patch ours emits is executed against the instructor's test cases first, so a defective rule degrades into abstention rather than into a wrong answer and the tool ships with no model behind it.

Three of these are close enough that their absence from our baselines needs stating. Refazer is the closest in spirit, since it also induces assignment-specific transformations, but it learns them from a student's own (before, after) edit, which neither corpus records. Verifix and the counterexample-guided repair of Orvalho, Janota, and Manquinho (2025) both operate on C, through MaxSMT over C types and a C-specific MaxSAT fault localizer, while our corpora are Python. We therefore take Refactory and PyDex as the reference-guided and per-submission LLM lines. Existing approaches, reference-guided and LLM-based alike, either fix the repair procedure by hand in advance or re-derive it from the model on every submission; neither leaves behind a reusable artifact for the assignment. To remedy this gap, we have the model author that artifact once per assignment, and repair every later submission with it and no model at all.

Approach

Task Definition

A programming assignment provides a problem description d, a set of test cases $T = \{t_1, \ldots, t_n\}$, and a set of submissions. Writing $p \models T$ when a program p passes every test case in T, the submissions divide into the buggy programs $B = \{p : p \not\models T\}$ and the correct programs $R = \{p : p \models T\}$. Since the correct program used to repair a buggy program is a reference program, R is also the pool of reference programs. Repairing the buggy programs produces a set of fixed programs F, with individual members denoted b, r, and f. To repair b is to produce an f with $f \models T$, and nothing weaker counts: an f passing only some test cases is not a repair.

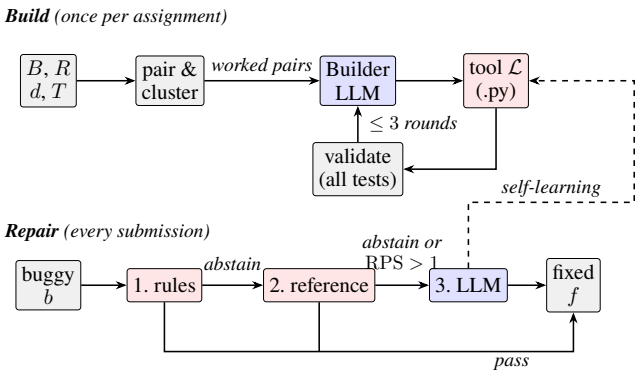


Figure 2: Overview of CoEP. LLM effort (blue) is spent once per assignment to build a local repair tool (red); at repair time the tool tries rules and a reference transplant first, so most submissions never reach a model. The dashed path is self-learning ($\text{CoEP}_{\text{L+SL}}$).

Overview

CoEP separates the two things a repair system does (Figure 2). The *build* happens once per assignment. A local step that costs no model call turns the class's own submissions into a small set of worked pairs, and a builder LLM turns those, together with d and T, into a *local repair tool* $\mathcal{L}$: an ordinary Python module with no model access. *Repair* happens on every submission and consults $\mathcal{L}$ first. Given a buggy program b, the tool tries three strategies in increasing order of cost: its own rewrite rules, a transplant from a reference program, and, only if both abstain or the local patch is larger than the program it repairs, a single conventional model call. Whichever strategy first produces a candidate that passes every test case in T supplies the fixed program f, so a model is involved in the third alone.

Three configurations follow from where we stop. CoEP_{L} uses the tool alone and spends nothing per submission beyond CPU time. CoEP_{LF} adds the model call for the submissions the tool cannot serve. $\text{CoEP}_{\text{L+SL}}$ adds self-learning: whenever the fallback repairs a submission the tool missed, the model generalizes that repair into a new named rule and the tool absorbs it, so an assignment's tool improves over a semester rather than staying frozen at whatever one build session produced.

Build: Local Repair Tool

Worked pairs. This step is local and costs no model call: the builder sees a *chosen* sample of B rather than all of it, one program per cluster, which on our assignments is 43–71% of the pool. Let $\sigma(p)$ collect the node types of p's AST read in breadth-first order, one sequence $\sigma_i(p)$ per top-level statement, and, with lev the Levenshtein distance between two such sequences, let $\delta(b, r)$ be the cost of the cheapest one-to-one matching M of b's sequences to r's, $\min_M [\sum_{(i,j)\in M} \mathrm{lev}(\sigma_i(b), \sigma_j(r)) + \sum_{i \notin M} |\sigma_i(b)| + \sum_{j \notin M} |\sigma_j(r)|]$, so a missing or surplus definition costs what it is. Every buggy program takes as partner the reference program it is fewest edits from, $\pi(b) = \arg\min_{r \in R} \delta(b, r)$, rather than the one that merely looks closest, and $\varepsilon(b)$ is the multiset of edit operations taking b to $\pi(b)$, refined to a value-level alignment when the node-type one comes out empty, since a wrong identifier or literal is invisible to node types alone.

We then partition $B = C_1 \uplus \cdots \uplus C_k$ by average linkage over cosine distance between TF-IDF vectors of the $\varepsilon(b)$, with k chosen to maximize mean silhouette. Both choices were measured rather than assumed: a rule induced from a representative pays off on the members needing the *same edit*, so grouping by shape fills the largest clusters with programs that merely look alike, and a fixed cut height is no smaller an assumption than a fixed k, one constant giving six-fold different granularity across our assignments. Each cluster contributes its *modal* repair, $b_i^\star = \arg\max_{b \in C_i} |\{b' \in C_i : \varepsilon(b') = \varepsilon(b)\}|$, degenerating to the medoid where no edit repeats. The medoid is the most typical *program* and usually a multi-issue tangle; the smallest edit is the submission that came closest to correct, an outlier rather than a spokesman. The build set is $S = \{(b_i^\star, \pi(b_i^\star))\}_{i=1}^k$, ordered by increasing δ, so a weak builder is shown each buggy program *with* a reference program that repairs it and need only generalize the transformation.

Rule authoring. The builder LLM turns (d, T, S), with T shown as up to eight example cases, into the assignment-specific tool $\mathcal{L}_0$: a module of AST rewrite rules, each named after the misconception it targets and tagged with the failure symptoms it addresses. A rule is a *candidate generator*, a map ρ from a program to a set of candidates, under a strict contract: it splices its edit as a source-span replacement rather than re-rendering the module, it proposes rather than claims, and it never rewrites a whole function. A rule that cannot express a targeted edit, that raises, or that runs past a two-second alarm yields $\rho(p) = \emptyset$ and the submission escalates.

Generate–validate–refine. A single-shot module can contain its own defects, so the harness closes the loop on the build set, refining from execution feedback (Madaan et al. 2023; Shinn et al. 2023). Write $\mathrm{cov}(\mathcal{L})$ for the fraction of S whose buggy half the rule search below repairs, $|\{(b, \cdot) \in S : \exists c \in \mathcal{L}^*(b),\ c \models T\}|/|S|$, with $\mathcal{L}^*(b)$ the candidate set that search enumerates; the transplant sits deliberately outside it, so a round is scored on the rules the

builder wrote and not on the substrate underneath them. Coverage is measured by *executing* T, never from the module's own claim. While $\mathrm{cov}(\mathcal{L}_j) < 1$ the harness returns the still-failing sources for revision, giving $\mathcal{L}_{j+1}$, for up to $r_{\max}$=3 builder calls; the frozen tool is $\mathcal{L} = \arg\max_{j \le r_{\max}} \mathrm{cov}(\mathcal{L}_j)$, ties going to the earliest round. Build cost is exactly the token count of these exchanges, measured from the raw strings.

Repair: Rules, Reference, LLM

Given b, the frozen tool $\mathcal{L}$ tries three strategies in increasing order of cost and returns the first candidate c it can verify with $c \models T$. The first two use no model; only the third spends a call.

Rule-based repair. The rules are composed by a symptom-ordered search over layers: $\Lambda_0 = \{b\}$ and $\Lambda_{j+1} = (\bigcup_{p \in \Lambda_j} \bigcup_{\rho \in \mathcal{L}} \rho(p)) \setminus \bigcup_{i \le j} \Lambda_i$, with b's failing-test signature ordering the ρ whose declared symptoms match. Every candidate in Λ_{j+1} is validated before Λ_{j+2} is enumerated, so the returned repair is the shallowest c with $c \models T$. A handful of always-on rules for the slips every class makes, comparison flips and off-by-one shifts, sit alongside whatever the builder writes, inverting the classical mutation-operator space (Just et al. 2014; Kim et al. 2013; Liu et al. 2019). Every surviving patch carries the ordered rule names behind it, provenance that aggregates into a per-class *misconception profile*.

Reference-based repair. Most student bugs need a structural edit no rule expresses; R supplies it. Let $H(b, r)$ be the differing line-hunks between b and one of the eight structurally nearest reference programs r, taken both raw and with r's variables renamed to the student's, parameters by position and locals by first binding, and let $b \oplus h$ be b with the hunks $h \subseteq H(b, r)$ applied. We validate the 300 smallest h with $1 \le |h| \le 3$ in increasing order of lines changed and return $\arg\min_h \mathrm{RPS}(b, b \oplus h)$ over the first five satisfying $b \oplus h \models T$, abstaining when none does, so the exact tree-edit score, the distance from b to $b \oplus h$ normalized by the size of b and defined with the other metrics in the Experimental Setup, is paid only on the handful that pass. This is reference reuse in the lineage of CLARA, Sarfgen, and Refactory (Gulwani, Radiček, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2020), at hunk granularity rather than the block granularity those systems use.

LLM-based repair. Conventional LLM-based APR maps every buggy program to a query $\mathrm{LLM}(d, T, b)$. CoEP keeps that call for the submissions the local strategies cannot serve, and its cost is counted against CoEP in every comparison. Writing f_{loc} for the local result, it fires when $f_{\mathrm{loc}} = \mathrm{fail}$ or $\mathrm{RPS}(b, f_{\mathrm{loc}}) > 1$. Abstention is not the only trigger because the acceptance test is the instructor's suite, which says nothing about patch size: the transplant can return a $c \models T$ larger than the student's whole program, while the rule layer must return $\emptyset$ whenever it cannot express a targeted edit. The threshold gives the transplant the same contract, since a patch larger than the program it repairs is a rewrite, not a repair, under any assignment's standards. The call is strengthened to match the student's strategy by attaching the minimum-edit reference program $r = \pi(b)$, the same pairing the build uses, and we keep $\arg\min_{f \in \{f_{\mathrm{loc}},\, \mathrm{LLM}(d,T,b,r)\},\ f \models T} \mathrm{RPS}(b, f)$, abstaining when neither passes, so escalating can cost a call but never coverage or patch size.

The threshold is a constant rather than a quantity calibrated per assignment. We chose it that way because its value is a definition and not a fitted number. Across the 29 settings we measured the repair rate is identical to four decimals and spend stays within \$12.47–\$15.61, so the choice carries no conclusion; the supplement plots them.

Self-learning. From an escalated pair (b, f) the LLM writes a new named rule ρ', kept only if $\exists\, c \in (\mathcal{L} \cup \{\rho'\})^*(b)$ with $c \models T$, in which case $\mathcal{L} \leftarrow \mathcal{L} \cup \{\rho'\}$. Absorption writes to a staging copy the evaluated pass never consults, so no submission's outcome depends on which preceded it, and the accumulated rules are a separate artifact, evaluated by a second frozen pass and priced to themselves. This is the $\mathrm{CoEP}_{\mathrm{L+SL}}$ configuration; it spends calls only on submissions that already escalated, so it too leaves the build claim untouched.

The first two strategies run on a CPU in $\sim$2–3 s per submission and offline. Under $\mathrm{CoEP}_{\mathrm{L}}$ no student code leaves the institution once the tool is built, though the build sends the worked pairs and $\mathrm{CoEP}_{\mathrm{LF}}$ sends each escalated submission: the guarantee belongs to the configuration, not the method. Rule application and transplant are themselves deterministic; the enumeration that feeds them is bounded by a wall clock, so under heavy load fewer candidates are examined and the tool may abstain where it would otherwise repair. A defective tool fails, never returning a wrong repair, because the harness re-validates every patch before a student sees it.

Experiments

Research Questions.

- **RQ1**: What is the cost impact of synthesizing a repair tool once per assignment, compared to querying an LLM once per submission?
- **RQ2**: How effective is CoEP in repairing programming assignments compared to state-of-the-art reference-guided and LLM-based repair approaches?
- **RQ3**: How does the LLM-authored rule layer impact the performance of CoEP?
- **RQ4**: How helpful is it to absorb the repairs of escalated submissions back into the tool as new rules?

Baselines. We compare CoEP against the two mainstream lines of program repair for programming assignments: (1) **Refactory** (Hu et al. 2020), which refactors the class's correct programs to match each buggy program's structure and transplants the divergent blocks; (2) **PyDex** (Zhang et al. 2024), which sends each buggy program to the model with the three most similar (buggy, correct) exemplars from the class pool, samples K=10 candidates, filters them by the test oracle, and returns the survivor closest to the student's program. We issue the ten as separate requests and bill all ten, where the original draws them from one; billing the prompt once would put PyDex at \$231 and the factor at 17 rather than 21.6. (3) $\mathbf{PyDex_{PC}}$ re-prices the same queries under *prompt caching*. A query reuses the key–value tensors computed for an identical leading prefix of an earlier one, billed

at roughly a tenth of the input rate (Anthropic 2025; OpenAI 2024). CoEP itself is evaluated in three configurations. **CoEP$_{\mathbf{L}}$** runs the frozen local tool alone with no deployment LLM use, accepting any patch the test oracle passes, which is the acceptance rule Refactory uses. **CoEP$_{\mathbf{LF}}$** adds the per-submission fallback on abstentions and on passing local patches above $\mathrm{RPS} = 1$. **CoEP$_{\mathbf{fw}}$** is the framework alone, the same tool with the builder's rules removed, which calls no model at any point. **CoEP$_{\mathbf{L+SL}}$** re-runs CoEP$_{\mathrm{L}}$ with the self-learned rules merged and frozen beforehand, so the arm is order-independent; its induction calls are priced to it.

Datasets. We evaluate on 11 problems and 2,970 buggy programs drawn from two real-student corpora: five CS1 assignments contributing 1,783 buggy and 2,442 correct programs at 7.4 test cases per assignment, and six Codeforces problems contributing 1,187 buggy and 698 correct at 35.5 test cases. The first five are a first-year university course's assignments with every submission its students wrote, released with Refactory (Hu et al. 2020) and since standard in this literature (Zhang et al. 2024). The other six come from Codeforces, whose submissions the open-r1 release mirrors (Penedo et al. 2025), sampled at ratings 800 through 2000 across five official bands (Codeforces 2024) and restricted to problems whose complete official test suites are public, so the oracle is the judge's own. Every submission a corpus labels buggy is a repair target for every arm, so the denominator is the pool as published. No train/test split is used, as prior per-assignment systems assume; the builder does see worked pairs drawn from the same buggy programs it is later measured on, 1,524 of the 2,970.

LLM Configuration. Every LLM arm runs on two models, one per provider: **gpt-5.4-mini** ($0.75/$4.50 per Mtok) and **claude-sonnet-5** ($2/$10 per Mtok). We select each provider's cheapest tier at which prompt caching is functional, in fairness to the caching baseline: PyDex$_{\mathrm{PC}}$ only competes if caching can engage, and we verified that the absolute cheapest tiers rule it out: Haiku 4.5 refuses to cache below its 4,096-token minimum, above these prompts, and GPT-5.4-nano's automatic caching did not engage on identical 1,200-token prompts, whereas both selected models cache at their documented 1,024-token minimums.

Evaluation Metrics. **Repair rate (RR)** is the percentage of buggy programs a system repairs, counted only when the fixed program passes *all* of the assignment's test cases. **Relative patch size (RPS)** is the tree-edit distance between the buggy and fixed ASTs normalized by buggy program size, reimplemented from Refactory's definition (Hu et al. 2020), and is averaged over what an arm repairs. **Average time taken** $\mathrm{ATT}_{\mathrm{tot}} = \mathrm{ATT}_{\mathrm{cpu}} + \mathrm{ATT}_{\mathrm{api}}$ is mean wall-clock seconds per submission, local search plus model latency. **LLM Cost** $C_{\mathrm{tot}} = C_{\mathrm{build}} + C_{\mathrm{api}}$ is US dollars over the whole benchmark, the one-off synthesis plus the per-submission inference, priced at the tiers above from exact token counts; a system with no build phase has $C_{\mathrm{build}} = 0$. The **escalation rate** e is the fraction of submissions taking a model call.

Setup. All experiments run on one Apple M1 Ultra workstation (64 GB RAM) under Python 3.13, CPU only but for the API calls, with no two submissions evaluated concurrently, so ATT is a measurement rather than a contention artifact. Every call is billed and timed; CoEP$_{\mathrm{L}}$ and CoEP$_{\mathrm{LF}}$ come from a single pass, and Refactory runs unchanged, ported to Python 3. Local search and evaluation are deterministic; the one stochastic step is builder synthesis, which we do not repeat, and the acceptance oracle bounds what a reseeded build could change: every candidate runs against the instructor's suite before it is returned, so another build moves which submissions the tool abstains on, not whether a returned patch passes. Code, the built tools, and every result file are in the submitted Code and Data Package.

Results

RQ1: Cost Effectiveness. Our objection to per-submission repair was structural: spending scales with enrollment and recurs every term, and a cheaper tier or a cache lowers the price of a call without changing how many calls are made. This question answers what moving the call to once per assignment costs, how many submissions repay the one-off build, and what the arrangement costs in turnaround. Figure 3 plots the cumulative C_{tot} one assignment spends against the submissions it receives, once per model: a per-submission arm is a line through the origin whose slope is the price of one query, while CoEP starts at C_{build} and rises only on the fraction e that escalates, so with Δ the marginal per-submission saving, itself averaged over escalating and non-escalating submissions, the two cross at $N^* = C_{\mathrm{build}}/\Delta$. The supplement splits turnaround into local search and the model wait behind it.

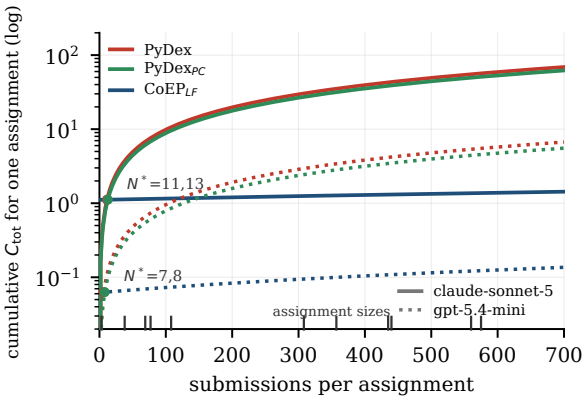


Figure 3: Cumulative C_{tot} for one assignment against its enrollment, solid for claude-sonnet-5 and dotted for gpt-5.4-mini. Dots mark where CoEP$_{\mathrm{LF}}$ overtakes each baseline; ticks along the axis are the eleven assignments' actual sizes, ten of which clear N^*.

Three findings follow. **(1) In dollars**, CoEP$_{\mathrm{LF}}$ spends $C_{\mathrm{tot}} = \$13.58$ over the whole benchmark on claude-sonnet-5 and $1.00 on gpt-5.4-mini, against PyDex's $292.77 and $28.42, **21.6 and 28.4 times** ours, a ratio that scales with the $K{=}10$ candidates PyDex draws per submission and would be near $2\times$ at $K{=}1$; prompt caching leaves PyDex$_{\mathrm{PC}}$ at $263.05 and $23.50, **still 19.4 and 23.5 times**, because it pays for a

shared leading *prefix* and the variable buggy program comes first. **(2) In break-even**, an assignment repays its build after $N^* = \mathbf{11}$ submissions against PyDex and 13 against PyDex_{PC}, on claude-sonnet-5, and after 7 and 8 on gpt-5.4-mini, where the build falls faster (17.8×) than the price of a query (10.3×) and break-even therefore arrives sooner. Ten of the eleven are past either point, most by an order of magnitude. **(3) In turnaround**, ATT_{tot} is 3.06 s against 14.64 s on claude-sonnet-5 but 3.22 s against 2.14 s on gpt-5.4-mini, the one measure a per-submission arm can win, since a small model answers faster than our search enumerates and re-executes candidates. PyDex_{PC} re-prices queries rather than re-issuing them, so its turnaround is PyDex's. What we pay for the first two is 22 minutes of build per assignment on claude-sonnet-5 (5.9–40.0), once and offline. Below N^* a per-query repairer is the right tool; above it, and since every arm runs at the cheapest tier where caching functions, the measured gaps are a *floor*.

RQ2: Repair Effectiveness. A repairer is judged on two things: how many submissions it fixes, and how much of the student's program it disturbs while fixing them. This question answers where a tool synthesized once per assignment stands on both, and how that varies across assignments. Table 1 reports per-assignment RR and RPS with the two builders pooled; RPS is averaged over what each arm repairs, so a low RPS beside a low repair rate means an easier subset rather than a better patch.

Three findings follow. **(1) Against the symbolic baseline**, the local tool alone repairs **99.1%** at mean RPS **0.463** where Refactory repairs 85.9% at 0.500, and its repair rate is at least Refactory's on every one of the eleven assignments. On Refactory's own corpus the same holds, 98.5% against its published 90.8% at 0.306 against 0.397. That comparison crosses instruments: 0.397 is the authors' figure, and measuring their tool the way we measure everyone gives 0.365, so the gap is 16% rather than 23%. Either way it costs no marginal LLM spend. **(2) Against per-submission LLM repair**, coverage is level: CoEP_{L} reaches 99.1% and PyDex 98.9%, and adding the escalation tail takes CoEP_{LF} to **99.7%**, at 100% on ten of the eleven assignments. What separates the arms is patch size, **0.463** against **0.634** and 0.410 for CoEP_{LF}, 27% and 35% smaller, on ten and on all eleven assignments respectively. **(3) The per-assignment rows** show what a corpus mean hides, and what predicts it. Refactory's rate swings from 40.3% to 100% while no arm that consults a model drops below 93.8%: grouped into bands, Refactory falls 90.8% on CS1 to 83.1% on Codeforces 800–1000 to **54.5%** on 1200–2000, while CoEP_{LF} holds 99.5/100.0/100.0. Thin pools alone do not explain that collapse, since C5 and C6 give every arm only two correct programs to work from and CoEP_{fw}, which keeps the transplant and drops the builder's rules, repairs both at 100%. What separates them is the granularity of reuse: Refactory adopts divergent blocks, so a pool that solves the problem another way leaves nothing adoptable, while we adopt line-hunks after renaming the reference program's variables to the student's, which stays applicable when the strategies differ. RPS is averaged over what each arm repairs, which could flatter an arm that repairs less, so we also compare on the **5,025** arm–submission pairs every arm repairs: Refactory 0.479, PyDex 0.626, CoEP_{L} **0.434** and CoEP_{LF} **0.394**. The ordering is unchanged. Overall, the synthesized tool beats the symbolic one on both coverage and patch size, matches per-submission LLM repair on coverage while disturbing a third less of the student's program, and does so on whichever builder writes it.

RQ3: Ablation Study. The previous two questions treated the tool as one object. This one opens it and asks what the model-authored rules add over the symbolic substrate beneath them. That substrate is CoEP_{fw}, a first-class arm in Table 1: the identical tool with the builder's rules removed, which calls no model at any point and is therefore one artifact under either builder. Figure 4 gives the decomposition it acts on.

Three findings follow. **(1) The reach comes from the symbolic substrate.** CoEP_{fw}, which never calls a model, repairs **98.69%** at mean RPS **0.492**: far above Refactory's 85.93%, and level on coverage with PyDex's 98.94% at a smaller patch than its 0.634. We report this rather than bury it: it is an unfavourable result for per-submission LLM repair, not for the reformulation. If 2,970 calls and $292.77 buy a quarter of a point and a larger patch over a substrate that costs no model spend, that spending is what we are objecting to. **(2) The authored layer buys the last stretch above it.** Adding it takes the repair rate to 99.06% and the mean patch to 0.463, and that 6% patch reduction is diluted across the 75% the transplant handles either way. On the **1,447** arm–submission pairs where a rule fires, 992 distinct programs, the rule's patch is RPS **0.100** against the substrate's **0.213** on the same programs, **2.1 times** larger without it, and on 13 of them the substrate produces no repair at all, though the fallback and PyDex both do. Labelling thins with it: repairs carrying a builder-written name fall from 17.8% of all repairs to none, and only the always-on rules' names remain, over 10.1%. We claim legibility for those names, not correctness: nothing here validates that a rule's name is the misconception a tutor would assign. **(3) The substrate's reach is borrowed from the pool.** Table 2 holds each tool fixed and varies only how many correct programs the transplant may draw on, over submissions no builder was shown, from ten assignments, with the claude-sonnet-5 tool. Down to one correct program the two arms are indistinguishable on coverage, while the patch gap widens from 3% on the whole pool to 13% at one. Take the pool away entirely and the substrate has nothing to transplant from: it repairs **11.6%** where the same tool with its rules repairs **32.2%**. That is a repair-time restriction and not a cold start: every tool here was built with the whole pool available, since its worked pairs are drawn from it. The rule layer is insurance against a thin pool, and a benchmark whose assignments each arrive with hundreds of correct submissions is precisely where that insurance looks worthless. Whether $1.11 per assignment is worth that is a departmental question; a course on a deep pool may reasonably decline. Our claim is not that the LLM produces the performance but that this is where LLM effort belongs.

The rules sort by generality, and generality predicts

	ID	Assignment	Rating	RR (%) Refactory	RR (%) PyDex	RR (%) CoEP_{fw}	RR (%) CoEP_{L}	RR (%) CoEP_{LF}	RPS Refactory	RPS PyDex	RPS CoEP_{fw}	RPS CoEP_{L}	RPS CoEP_{LF}
Refactory	R1	Sequential search	–	99.0	99.7	100.0	100.0	100.0	0.390	0.428	0.243	0.225	0.222
	R2	Unique dates/months	–	78.2	97.8	91.3	93.8	97.9	0.560	0.428	0.428	0.398	0.394
	R3	Duplicate elimination	–	97.4	99.8	100.0	100.0	100.0	0.340	0.423	0.284	0.280	0.280
	R4	Sorting tuples	–	88.2	99.3	100.0	100.0	100.0	0.310	0.535	0.388	0.386	0.291
	R5	Top-k elements	–	88.0	95.4	100.0	100.0	100.0	0.320	0.244	0.196	0.193	0.191
Codeforces	C1	Nearly Lucky Number	800	85.2	99.9	100.0	100.0	100.0	0.588	0.685	0.616	0.584	0.494
	C2	cAPS lOCK	1000	81.4	97.9	100.0	100.0	100.0	0.724	1.167	0.841	0.762	0.648
	C3	Planning Expedition	1200	47.8	100.0	98.6	98.6	100.0	0.847	0.705	0.640	0.638	0.585
	C4	Measuring Lengths	1400	92.1	100.0	100.0	100.0	100.0	0.605	0.643	0.531	0.571	0.525
	C5	Jumping Jack	1600	40.3	100.0	100.0	100.0	100.0	0.878	0.914	0.950	0.914	0.826
	C6	Digital Root	2000	100.0	100.0	100.0	100.0	100.0	2.248	1.975	1.777	1.652	1.652
	Overall			**85.9**	**98.9**	**98.7**	**99.1**	**99.7**	**0.500**	**0.634**	**0.492**	**0.463**	**0.410**

Table 1: Overall repair effectiveness on 11 assignments, with the two builders pooled. **RR** is the repair rate and **RPS** the relative patch size. CoEP_{fw} is the framework alone, the substrate with the builder's rules removed, and uses no model at any point. Refactory's CS1 rows are its published figures; its Codeforces rows are our run of the released tool.

	RR (%)		RPS	
pool	CoEP_{fw}	full	CoEP_{fw}	full
0	11.6	32.2	0.072	0.194
1	97.9	97.9	1.113	0.963
2	99.5	99.5	0.906	0.811
5	99.8	99.8	0.683	0.624
all	99.3	99.3	0.482	0.467

Table 2: Repair rate and patch size as the transplant's reference pool is restricted to k correct programs, over 146 submissions no builder was shown. Each sampled pool is drawn three times and the entry is the mean; the spread across draws never exceeds 2.1 points of RR or 0.405 RPS.

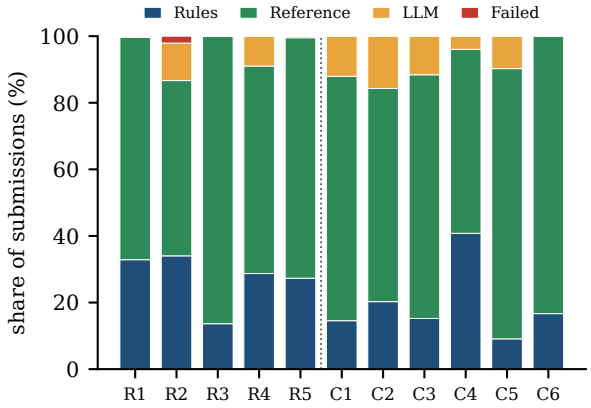


Figure 4: Decomposition of the ablation.

reach. `add_missing_call_parens` names a Python trap that belongs to no assignment and is the only one of 128 to fire on two; `slice_case_check_to_equality` names C2's misconception and repairs 108 of its submissions; `carry_fix_two_value_print` is tied to C4's problem statement and repairs 28 of its 38 alone. That the layer is assignment-specific is a consequence of level, not a defect.

RQ4: Self-Learning. The frozen tool leaves a record of what it could not repair. When the fallback then repairs one of those misses, the model generalizes that fix into a named rule, and the rule is kept only if the augmented tool repairs the miss on its own. This question answers whether such a rule holds outside the submission it was induced from. Absorption writes to a staging copy the evaluated pass never consults and the augmented tool is frozen before this arm runs, so the comparison is order-independent; the supplement plots the funnel.

Two findings follow. **(1) The material is scarce and the bar is high.** CoEP_{L} fails on 56 of 5,940 submissions, the fallback repairs 38, and those 38 attempts leave **8** rules. That is not a failure of the mechanism but what a 99.06% local repair rate means. **(2) The rules that survive generalize past the submissions that induced them.** All eight went on to repair, recovering **21** submissions between them, and since eight submissions induced them, **13** of those 21 are programs the rules had never seen. One submission regressed, so the net is 20, and the induction cost $0.76 priced to the artifact. Self-learning fires only where the tool failed and such places are rare, but where it fires the rules carry past their own examples, for under a dollar per assignment.

Conclusion

We reformulate program repair for programming assignments so that the model writes the repair *tool* rather than the patch, once per assignment; the frozen tool then repairs with no model at all. On 2,970 student submissions it beats a reference-guided repairer on coverage and patch size, and matches per-submission LLM repair at 21.6 times lower cost at PyDex's candidate budget. The model buys smaller patches and a named misconception where pools are deep, and the reach itself where they are not.

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